

COMP 421: Files & Databases

Lecture 3: Storage Manager

Logistics

2

Project 0 Postmortem: less than 80/100, come talk to me

Office hours schedule posted!

Project 1 is live! (8/31)

Sneak Preview

- P1: Buffer Pool Manager
- P2: B+ Tree
- P3: Query Executor / Optimizer

Bootcamp 2 RSVP (week of 9/8)

Last Class

We now understand what a database looks like at a logical level and how to write queries to read/write data (e.g., using SQL).

We will next learn how to build software that manages a database (i.e., a DBMS).

Course Outline

Relational Databases

➔ Storage

Query Execution

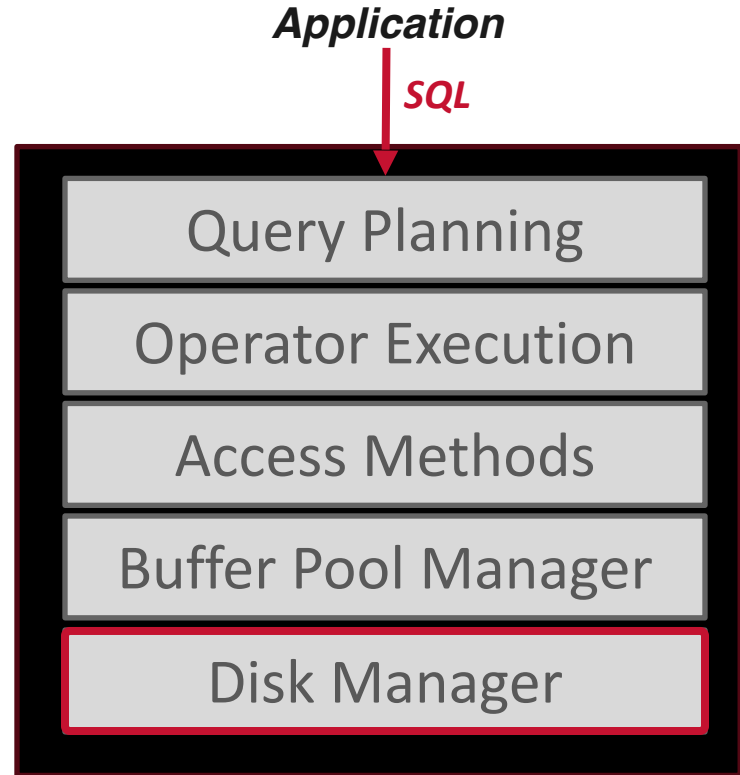
Concurrency Control

Database Recovery

Distributed Databases

Potpourri

DBMS



Today's Agenda

Background

File Storage

Page Layout

Tuple Layout

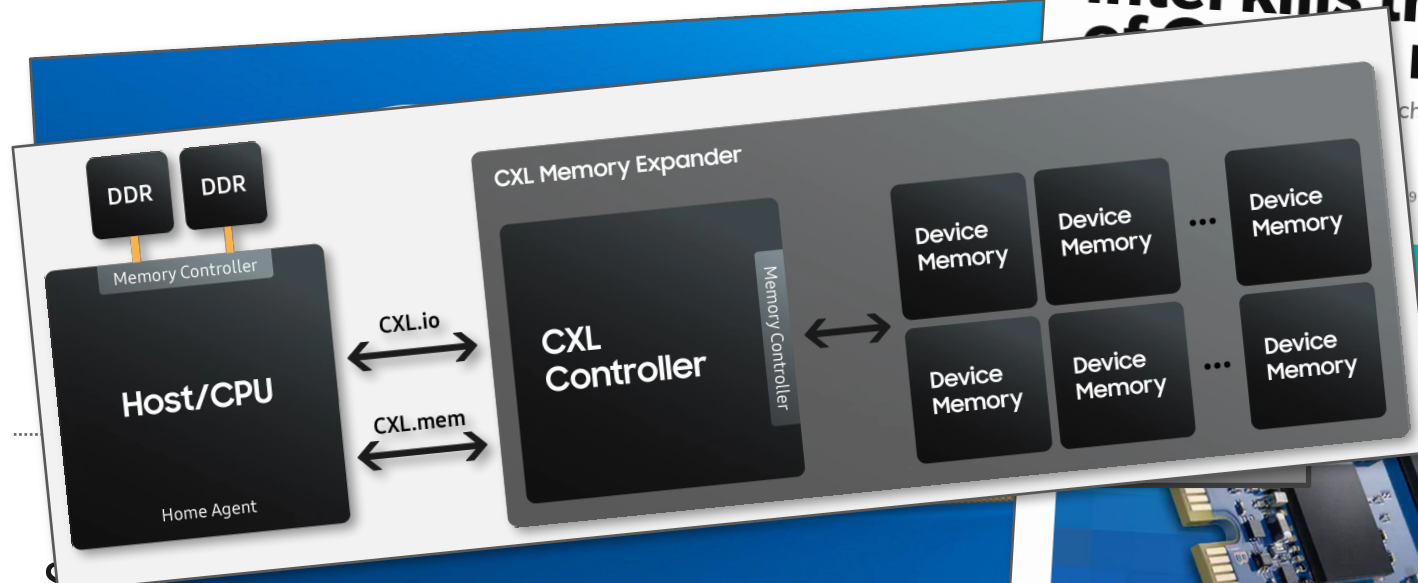
Disk-based Architecture

The DBMS assumes that the primary storage location of the database is on non-volatile disk.

For computation, data needs to be brought into volatile memory.

The DBMS's components manage the movement of data between storage and memory.

Storage Hier



Disk

Network



If you haven't built a super-high-end workstation in a while, you might not have heard of [Intel's Optane memory caching tech](#). Optane also powered ultra-fast SSDs for consumers and businesses alike. Not that it matters much now. After a disastrous second-quarter earnings call in which it missed expected revenue by billions of dollars, the company announced its plans to end its Optane memory business entirely.

Access Times

Latency Numbers Every Programmer Should Know

1 ns	L1 Cache	← 1 sec
4 ns	L2 Cache	← 4 sec
100 ns	DRAM	← 100 sec
16,000 ns	SSD	← 4.4 hours
2,000,000 ns	HDD	← 3.3 weeks
~50,000,000 ns	Network Storage	← 1.5 years
1,000,000,000 ns	Tape Archives	← 31.7 years

Sequential Vs. Random Access

Random access on non-volatile storage is almost always much slower than sequential access.

DBMS will want to maximize sequential access.

- Algorithms try to reduce number of writes to random pages so that data is stored in contiguous blocks.
- Remember: random vs. sequential $\approx 100x$ difference!

Reads and Writes

Other Numbers Every Programmer Should Know

1 ns	L1 Cache Ref	
4 ns	L2 Cache Ref	Byte addressable (64-bit word)
100 ns	DRAM	
16,000 ns	SSD	
2,000,000 ns	HDD	4KB block (4-16 KB phys. page)
~50,000,000 ns	Network Storage	
1,000,000,000 ns	Tape Archives	4KB block (128 MB block, HDFS)

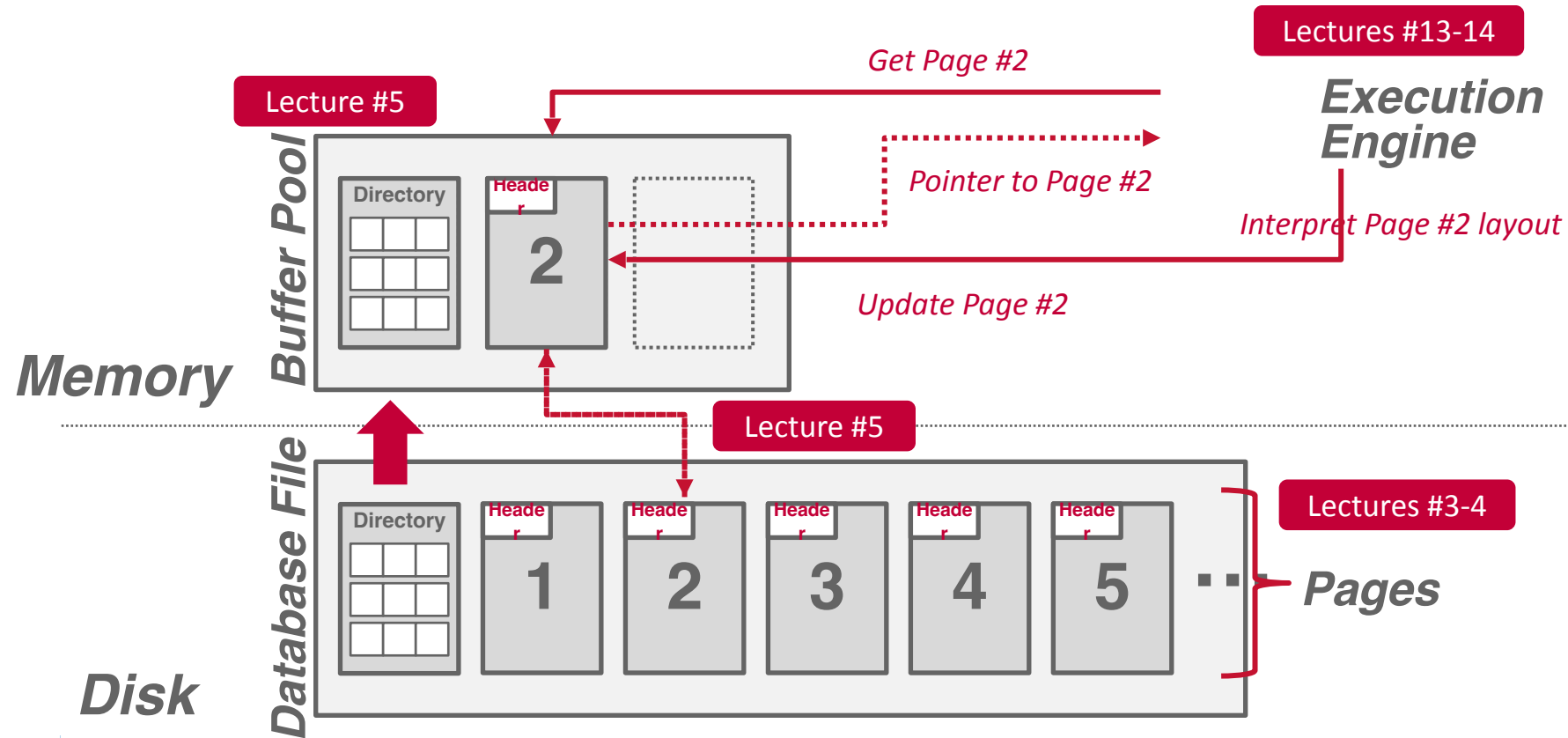
System Design Goals

Reading/writing to disk is expensive, so it must be managed carefully to avoid large stalls and performance degradation.

Random access on disk is usually much slower than sequential access, so the DBMS will want to maximize sequential access.

Allow the DBMS to manage databases that exceed the amount of memory available.

Disk-oriented DBMS



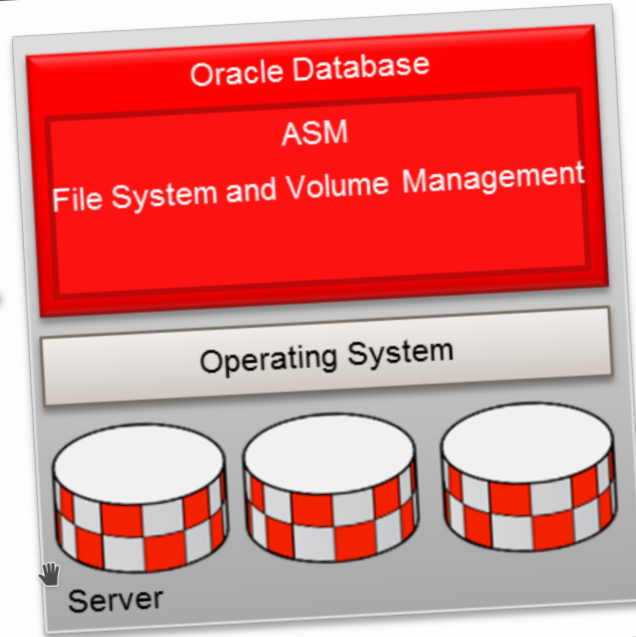
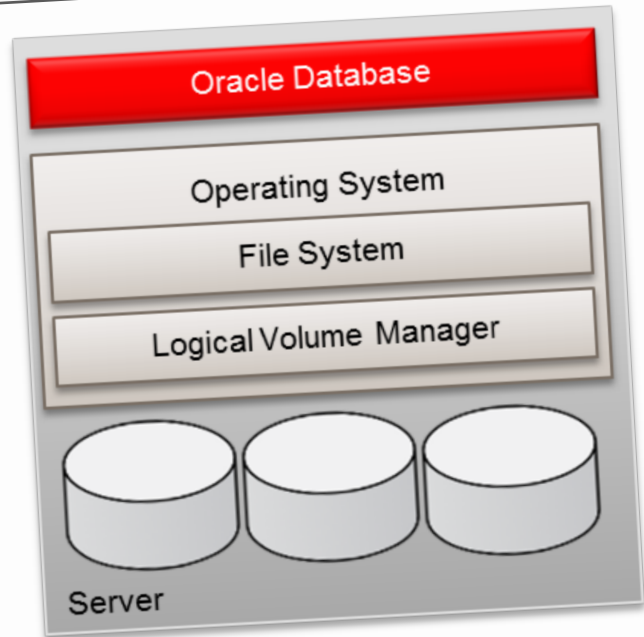
Database Storage

Problem #1: How the DBMS represents the database in files on disk.

← **Today**

Problem #2: How the DBMS manages its memory and moves data back-and-forth from disk.

File Storage



Storage Manager

The storage manager is responsible for maintaining a database's files.

- Some do their own scheduling for reads and writes to improve spatial and temporal locality of pages.

It organizes the files as a collection of pages.

- Tracks data read/written to pages.
- Tracks the available space.

A DBMS typically does not maintain multiple copies of a page on disk.

- Assume this happens above/below storage manager.

Database Pages

A page is a fixed-size block of data.

- It can contain tuples, meta-data, indexes, log records...
- Most systems do not mix page types.
- Some systems require a page to be self-contained.

Each page is given a unique identifier (**page ID**).

- A page ID could be unique per DBMS instance, per database, or per table.
- The DBMS uses an indirection layer to map page IDs to physical locations.

Database Pages

There are three different notions of "pages" in a DBMS:

- Hardware Page (usually 4KB)
- OS Page (usually 4KB, x64 2MB/1GB)
- Database Page (512B-32KB)

A hardware page is the largest block of data that the storage device can guarantee failsafe writes.

DBMSs that specialize in read-only workloads have larger page sizes.

Default DB Page Sizes

4KB

8KB

16KB



Page Storage Architecture

Different DBMSs manage pages in files on disk in different ways.

- Heap File Organization
- Tree File Organization
- Sequential / Sorted File Organization (ISAM)
- Hashing File Organization

At this point in the hierarchy, we do not need to know anything about what is inside of the pages.

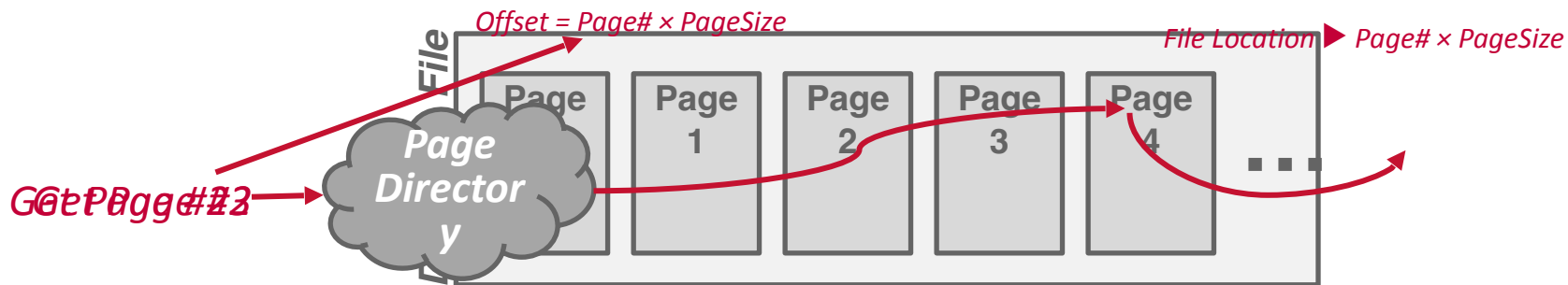
Heap File

A heap file is an unordered collection of tuples, tuples are stored on pages in random order.

→ Create / Get / Write / Delete Page

→ Must also support iterating over all pages.

Need additional meta-data to track location of files and free space availability.



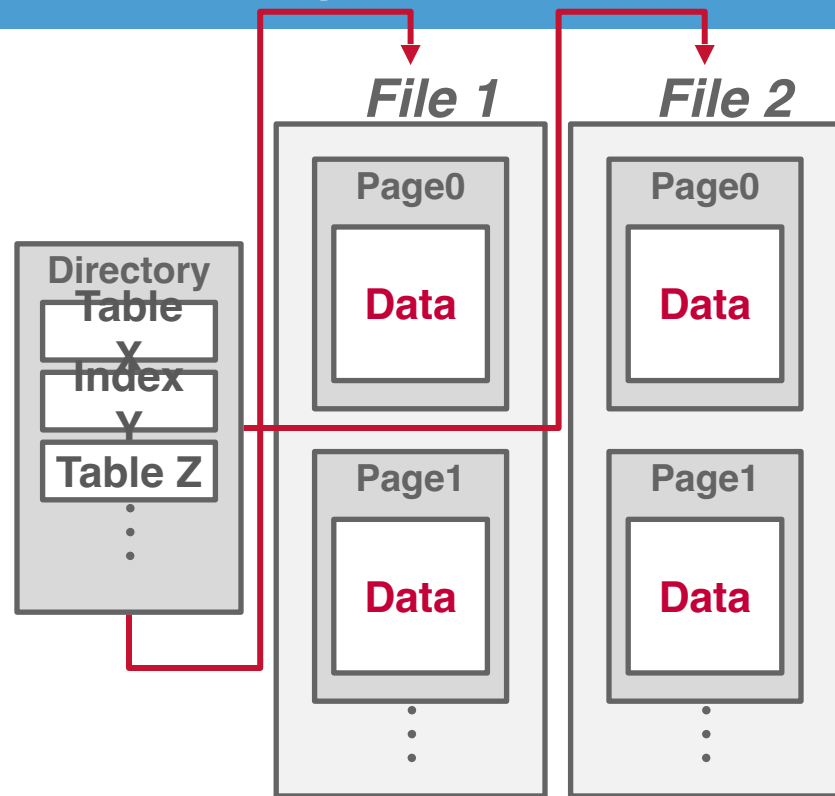
Heap File: Page Directory

The DBMS maintains special pages that tracks the location of data pages in the database files.

- One entry per database object.
- Must make sure that the directory pages are in sync with the data pages.

DBMS also keeps meta-data about pages' contents:

- Amount of free space per page.
- List of free / empty pages.
- Page type (data vs. meta-data).



Today's Agenda

File Storage

Page Layout

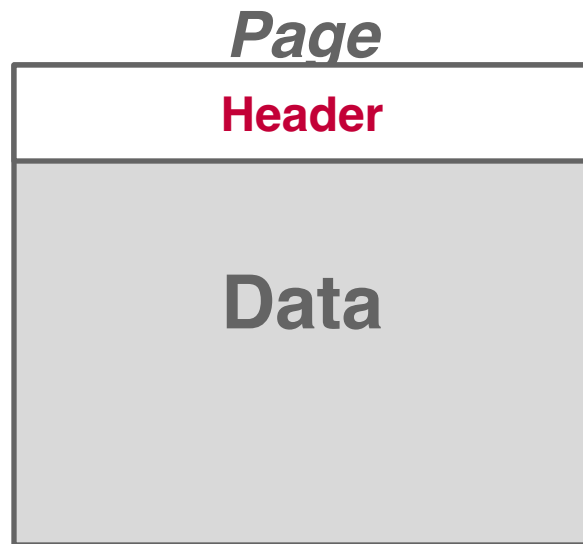
Tuple Layout

Page Header

Every page contains a header of meta-data about the page's contents.

- Page Size
- Checksum
- DBMS Version
- Transaction Visibility
- Compression / Encoding Meta-data
- Schema Information
- Data Summary / Sketches

Some systems require pages to be self-contained (e.g., Oracle).



Page Layout

For any page storage architecture, we now need to decide how to organize the data inside of the page.

→ We are still assuming that we are only storing tuples in a

Lecture #6

row-oriented storage model.

Approach #1: Tuple-oriented Storage

← **Today**

Approach #2: Log-structured Storage

Approach #3: Index-organized Storage

Lecture #4

Tuple-oriented Storage

How to store tuples in a page?

Strawman Idea: Keep track of the number of tuples in a page and then just append a new tuple to the end.

- What happens if we delete a tuple?
- What happens if we have a variable-length attribute?

<i>Page</i>	
Num Tuples = 2	
Tuple #1	
Tuple #4	
Tuple #3	

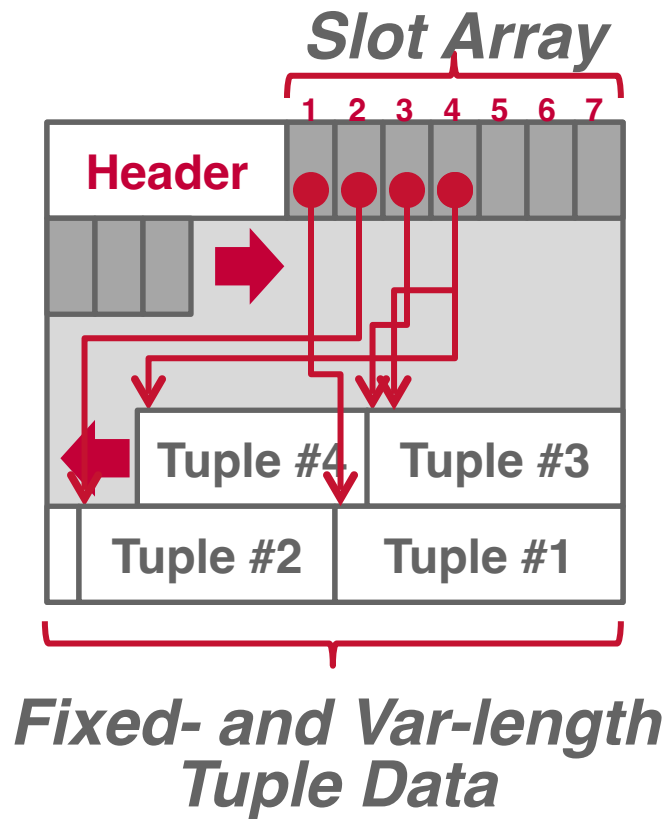
Slotted Pages

The most common layout scheme is called slotted pages.

The slot array maps "slots" to the tuples' starting position offsets.

The header keeps track of:

- The # of used slots
- The offset of the starting location of the last slot used.



Record IDs

The DBMS assigns each logical tuple a unique record identifier that represents its physical location in the database.

- File Id, Page Id, Slot #
- Most DBMSs do not store ids in tuple.
- SQLite uses [ROWID](#) as the true primary key and stores them as a hidden attribute.

Applications should never rely on these IDs to mean anything.

CTID (6-bytes)

ROWID (8-bytes)

%%physloc%% (8-bytes)

ROWID (10-bytes)

Today's Agenda

File Storage

Page Layout

Tuple Layout

Tuple Layout

A tuple is essentially a sequence of bytes.

→ These bytes do not have to be contiguous.

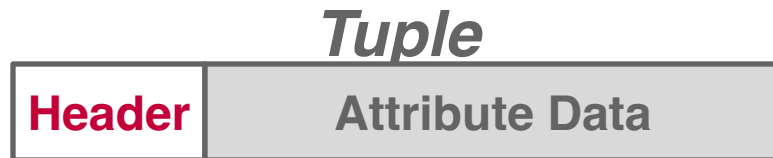
It is the job of the DBMS to interpret those bytes into attribute types and values.

Tuple Header

Each tuple is prefixed with a header that contains meta-data about it.

- Visibility info (concurrency control)
- Bit Map for **NULL** values.

We do not need to store meta-data about the schema.



Tuple Data

Attributes are typically stored in the order that you specify them when you create the table.

This is done for software engineering reasons (i.e., simplicity).

However, it might be more efficient to lay them out differently.

Tuple

Header	a	b	c	d	e
--------	---	---	---	---	---

```
CREATE TABLE foo (  
  a INT PRIMARY KEY,  
  b INT NOT NULL,  
  c INT,  
  d DOUBLE,  
  e FLOAT  
);
```

Denormalized Tuple Data

DBMS can physically *denormalize* (e.g., "pre-join") related tuples and store them together in the same page.

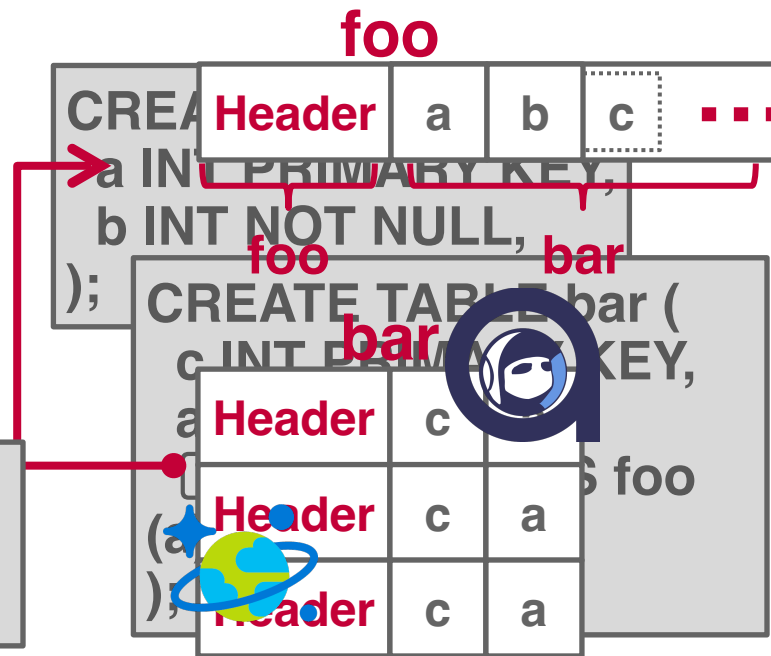
- Potentially reduces the amount of I/O for common workload patterns.
- Can make updates more expensive.

Not a new technique

→ IBM System/360

→ Several MySQL versions calling it "pre-join"

```
SELECT * FROM foo JOIN bar
ON foo.a = bar.a;
```



Conclusion

Database is organized in pages.

Different ways to track pages.

Different ways to store pages.

Different ways to store tuples.

Next Class

Log-Structured Storage
Index-Organized Storage
Value Representation
Catalogs